Practice Problems on Divisive Hierarchical Clustering

1D and 2D Datasets | Distance-Based Splitting and MST Approach

Concept: Divisive clustering is a top-down hierarchical clustering technique. All data points begin in one cluster, and the cluster is repeatedly divided into smaller clusters until the required number of clusters is obtained. The split rule must be specified; the problems below use the rules stated in each question.

Part A: Problems Based on 1D Dataset

Problem 1: Divisive Clustering Using a 1D Dataset

Question: Perform divisive hierarchical clustering on X = {2, 3, 4, 10, 11, 12}. Divide the sorted data at the largest gap between consecutive points.

Tasks

1. Arrange the data points in ascending order.
2. Calculate the distances between consecutive points.
3. Identify the largest distance and divide the dataset into two clusters.
4. Further divide the clusters until four clusters are obtained.
5. Draw the hierarchical clustering tree (dendrogram).

Worked solution

Step 1: The data are already sorted: X = {2, 3, 4, 10, 11, 12}.

Step 2: For 1D data, the distance between two values is $d(x_i, x_j) = |x_i - x_j|$.

Consecutive points	Distance
2 and 3	$\|2 - 3\| = 1$
3 and 4	$\|3 - 4\| = 1$
4 and 10	$\|4 - 10\| = 6$
10 and 11	$\|10 - 11\| = 1$
11 and 12	$\|11 - 12\| = 1$

Step 3: The largest gap is 6, between 4 and 10. Therefore, the first split is:

Cluster 1: {2, 3, 4} **Cluster 2:** {10, 11, 12}

Step 4: Split each group further. Since the gaps within each group are tied, either gap may be selected. One valid four-cluster result is:

{2, 3}, {4}, {10, 11}, {12}

Problem 2: Additional 1D Practice

Question: Apply divisive clustering to X = {1, 2, 3, 8, 9, 15, 16}, using the largest gap between consecutive sorted points.

1. Calculate all consecutive distances.
2. Identify the first split and write the two clusters.
3. Continue splitting to obtain three clusters.
4. Continue splitting to obtain four clusters.

5. Draw the clustering tree.

Part B: Problems Based on 2D Dataset

Euclidean distance formula

For points $P(x_1, y_1)$ and $Q(x_2, y_2)$:

$$d(P, Q) = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$$

Problem 3: Divisive Clustering Using a 2D Dataset

Question: Perform divisive hierarchical clustering on the points below using Euclidean distance. Use a distance-based splitting method and explain the split rule you apply.

Point	X-coordinate	Y-coordinate
A	1	1
B	2	1
C	2	2
D	8	8
E	9	8
F	9	9

1. Calculate the Euclidean distances between relevant pairs of points.
2. Identify the two natural groups of points.
3. Divide the dataset into two clusters.
4. Further divide it into four clusters.
5. Draw the hierarchical clustering tree.

Hint: A, B and C are close to one another; D, E and F are also close to one another. Use the distance calculations to justify the separation between these groups.

Problem 4: Additional 2D Practice

Question: Apply divisive hierarchical clustering to the following points using Euclidean distance.

Point	X-coordinate	Y-coordinate
P	1	2
Q	2	3
R	3	2
S	7	8
T	8	9
U	9	8

1. Calculate the Euclidean distances between relevant pairs.
2. Identify the largest separation between groups.
3. Divide the dataset into two clusters.
4. Continue splitting until four clusters are obtained.
5. Draw the dendrogram and show the clusters at each stage.

Part C: Challenge Problem — MST-Based Divisive Clustering

Problem 5: Divisive Clustering Using a Minimum Spanning Tree (MST)

Question: Perform divisive hierarchical clustering on the following 2D points using the MST approach and Euclidean distance.

Point	X-coordinate	Y-coordinate
A	1	1
B	2	2
C	3	1
D	7	7
E	8	8
F	9	7

Tasks

1. Calculate Euclidean distances between relevant pairs of points.
2. Construct the Minimum Spanning Tree (MST).
3. Identify and remove the longest edge in the MST to obtain two clusters.
4. Remove additional selected edges to obtain four clusters.
5. Draw the MST before and after removing the edges.
6. Write the final clusters.

MST procedure

1. Construct the weighted graph using Euclidean distances.
2. Find the Minimum Spanning Tree.
3. Remove the longest edge in the MST to split the data into two clusters.
4. Remove additional edges as needed to obtain the required number of clusters.

Important: MST-based divisive clustering is different from splitting sorted 1D data at its largest consecutive gap. In 2D, clearly state the splitting rule used.

Quick Revision: 1D vs. 2D Divisive Clustering

Feature	1D Dataset	2D Dataset
Representation	One numerical value	X and Y coordinates
Distance	Absolute difference	Euclidean distance
Example	{2, 3, 4, 10, 11, 12}	{(1,1), (2,1), (8,8)}
Possible split rule	Largest consecutive gap	Distance-based or MST-based
Output	Groups of numerical values	Groups of coordinate points

Exam tip: Practise Problems 1 and 2 for 1D data, Problems 3 and 4 for 2D data, and Problem 5 for the MST-based method. Always show the distance calculations, the split at each stage, and the final clusters.